CHAPTER 1

DISTRIBUTED MESSAGING QUEUE

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Examine the design of a distributed messaging queue, addressing requirements, design considerations, and performance evaluations.

1 System Design: The Distributed Messaging Queue

1.1 What is a messaging queue?

A **messaging queue** is an intermediate component that connects the interacting entities, known as **producers** and **consumers**.
The producer *produces* messages and places them in the queue, while the *consumer* retrieves the messages from the queue and processes them. There may be multiple producers and consumers interacting with the queue simultaneously.
Here is an illustration of two applications interacting via a single messaging queue:

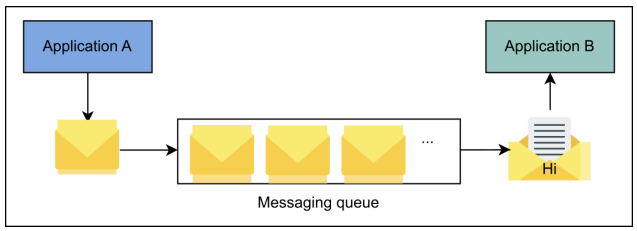


Figure 1.1: An example of two applications interacting via a single messaging queue

Key components of a messaging queue

The key components of messaging queues include:

Producers These are entities or applications that create and send messages to the queue. Producers generate data or events that need to be processed and push them into the messaging system for later consumption.

Consumers Consumers are entities or applications that receive and process messages from the queue. They subscribe to the queue and pull messages for processing, allowing them to handle tasks asynchronously and independently from the producers.

Queues A queue is a data structure that temporarily holds messages sent by producers until they are consumed by consumers. Queues ensure that messages are stored in a first-in, first-out (FIFO) order, although some systems may implement different ordering mechanisms.
They provide a buffer that decouples producers and consumers, allowing for smoother communication.

Messages Messages are the data packets that are sent from producers to consumers via the queue.
Each message typically contains a payload (the actual data) and metadata (such as headers or properties) that provide context about the message, such as its type, priority, or routing information. Together , these components enable efficient and reliable asynchronous communication in distributed systems, allowing for better resource management and improved system performance.
Let's discuss what advantages a messaging queue offers.

1.2 Why use a messaging queue

A messaging queue has several advantages and use cases.

- Messaging queues improve performance, reliability, scalability, and flexibility in distributed systems by enabling asynchronous, decoupled communication. Producers can send messages without waiting for consumers, reducing latency and preventing bottlenecks. Queues buffer messages during spikes or when consumers are offline, ensuring no data is lost and smoothing out load variations.

- They also decouple services, allowing each component to operate, scale, deploy, and fail independently. This independence simplifies maintenance and fosters more agile development. Queues further support load balancing by distributing work across multiple consumers, automatically routing messages to available workers, and maintaining throughput even when some consumers slow down or fail.

- Messaging systems enhance fault tolerance through persistence, retries, acknowledgments, and dead-letter queues, ensuring reliable delivery and easier troubleshooting. They make horizontal scaling straightforward—new consumers can be added at any time to increase processing capacity.

- Additional benefits include rate limiting, where queues absorb bursts of traffic to protect downstream services, and priority handling, which ensures that critical work is processed first through multiple queues or priority rules.

Overall, messaging queues provide a resilient, scalable backbone for modern applications, enabling smooth communication and consistent performance under varying loads.

Common messaging queue patterns

Point-to-point is a *one-to-one* messaging pattern where a producer sends a message to a single consumer via a queue.
Each message is delivered to exactly one designated consumer, making it ideal for tasks that must be handled by a specific worker. The queue tracks consumption and acknowledgments, ensuring guaranteed delivery. This model is commonly used in task queues and worker-based systems that need predictable, isolated processing.
Publish/subscribe (pub/sub or pub-sub) is a *one-to-many* pattern where a publisher sends messages to all subscribers interested in a topic, without knowing who they are.
Because producers and consumers are decoupled, the system scales easily as subscribers join or leave. Publish /subscribe is widely used in event-driven architectures for real-time updates (such as newsfeeds and stock data) and notification systems where messages must reach multiple consumers simultaneously.

Figure 1.2: Pub/Sub

Request/reply supports *synchronous, two-way* communication: a client sends a request and waits for a response. It 's used when immediate feedback is required, such as APIs, microservices, and transactional operations. Examples include checking product availability or orchestrating multiple services to fulfill a user request.
This pattern improves user experience by providing timely, coordinated responses.

Popular messaging queue technologies

While the concept is universal, several popular technologies implement these principles, each with different strengths:

- **RabbitMQ:** A widely used open-source message broker that supports multiple protocols (like AMQP) and complex routing patterns.

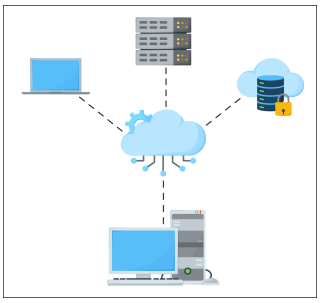


Figure 1.3: Point-to-point communication

- **Apache Kafka:** A high-throughput, distributed streaming platform often used for real-time data pipelines, event sourcing, and log aggregation.
- **Amazon Simple Queue Service (SQS):** A fully managed message queuing service from AWS, offering reliable, scalable queues that integrate easily with other cloud services.

Now, let's discuss the motivation behind designing a message queue.

Messaging queue use cases

A messaging queue has many use cases, both in single-server and distributed environments. For example, it can be used for interprocess communication within one operating system. It also enables communication between processes in a distributed environment.
Some of the use cases of a messaging queue are discussed below.

1. **Sending many emails:** Emails are used for numerous purposes, including sharing information, account verification, password resets, marketing campaigns, and more. All of these emails, written for different purposes, don't need immediate processing and, therefore, don't disturb the system's core functionality. A messaging queue can help coordinate a large number of emails between different senders and receivers in such cases.

2. **Data post-processing:** Many multimedia applications require processing content to meet the needs of different viewers, such as those for consumption on mobile phones and smart televisions. Oftentimes, applications upload the content into a store and use a messaging queue for post-processing of content offline. Doing this substantially reduces client-perceived latency and enables the service to schedule offline work at an appropriate time, probably late at night when compute capacity is less busy.

3. **Recommender systems:** Some platforms utilize recommender systems to deliver personalized content or information to users. The recommender system takes the user's historical data, processes it, and predicts relevant content or information. Since this is a time-consuming task, a messaging queue can be incorporated between the recommender system and requesting processes to increase and quicken performance.

1.3 Best practices for implementing messaging queues

Message idempotency

Message idempotency ensures that a message can be processed multiple times without causing duplicate actions or inconsistent data.
This is particularly essential in distributed systems, where retries, failures, or network issues may result in duplicate deliveries. Idempotent design guarantees that reprocessing produces the same outcome as a single execution, simplifying error recovery and protecting data integrity.
Overall, it strengthens system reliability and reduces the risk of unintended side effects.

Monitoring and logging

Monitoring and logging provide visibility into message flow and system health.
Monitoring helps teams detect bottlenecks, latency spikes, and failures in real time, enabling proactive intervention. Logging captures key events and errors for auditing, troubleshooting, and determining the root cause of issues.
Together, they improve operational transparency, support informed decisions, and enhance the resilience of messaging systems.

Figure 1.4: Error Handling

Error handling

Robust error handling ensures message processing continues smoothly even when failures occur.
Common techniques include automatic retries for transient issues, dead-letter queues to isolate messages that repeatedly fail, and circuit breakers to prevent cascading failures. Combined with strong logging and monitoring, these strategies help teams diagnose problems, protect data, and maintain reliable system behavior.

1.4 How do we design a distributed messaging queue?

We divide the design of a distributed messaging queue into the following five lessons:

1. **Requirements:** In this lesson, we focus on the functional and non-functional requirements of designing a distributed messaging queue. We also discuss a single-server messaging queue and its drawbacks in this lesson.

2. **Design Consideration:** In this lesson, we discuss several important factors that may affect the design of a distributed messaging queue, including the order of placing messages in a queue, their extraction, visibility in the queue, and the concurrency of incoming messages.

3. **Design:** In this lesson, we examine the design of a distributed messaging queue in detail. We also describe the process of replication of queues and the interaction between various building blocks involved in the design.

4. **Evaluation:** In this lesson, we evaluate the design of a distributed messaging queue based on its functional and non-functional requirements.

5. **Quiz:** At the end of the chapter, we assess your understanding of the design of a distributed message queue through a quiz.

Let's start by understanding the requirements of designing a distributed messaging queue.

2 Requirements of a Distributed Messaging Queue's Design

2.1 Requirements

In a **distributed messaging queue**, data resides on several machines. Our aim is to design a distributed messaging queue that has the following functional and non-functional requirements.

Functional requirements

Listed below are the actions that a client should be able to perform:

- **Queue creation:** The client should be able to create a queue and set some parameters—for example, queue name, queue size(A queue size is the maximum number of messages a queue can contain. For example, Amazon's standard queue size is unlimited. However, for inflight messages, the size is limited to 120,000.), and maximum message size. MaxMessageSize
- **Send message:** Producer entities should be able to send messages to a queue that's intended for them.
- **Receive message:** Consumer entities should be able to receive messages from their respective queues.
- **Delete message:** The consumer processes should be able to delete a message from the queue after a successful processing of the message.
- **Queue deletion:** Clients should be able to delete a specific queue.

Non-functional requirements

Our design of a distributed messaging queue should adhere to the following non-functional requirements:

- **Durability**: The data received by the system should be durable and shouldn't be lost. Producers and consumers can fail independently, and a queue with data durability is critical to make the whole system work, because other entities are relying on the queue.
- **Scalability**: The system needs to be scalable and capable of handling the increased load, queues, producers, consumers, and the number of messages. Similarly, when the load reduces, the system should be able to shrink the resources accordingly.
- **Availability**: The system should be highly available for receiving and sending messages. It should continue operating uninterrupted, even after the failure of one or more of its components.
- **Performance:** The system should provide high throughput and low latency.

2.2 Single-server messaging queue

Before we embark on our journey to map out the design of a distributed messaging queue, we should recall how queues(Such a queue is also called a bounded queue, due to its fixed size within the system.) are used within a single server where the producer and consumer processes are also on the same node. A producer or consumer can access a single-server queue by acquiring the locking mechanism to avoid data inconsistency. The queue is considered a critical section where only one entity, either the producer or consumer, can access the data at a time.
However, several aspects restrain us from using the single-server messaging queue in today's distributed systems paradigm. For example, it becomes unavailable to cooperating processes, producers and consumers, in the event of hardware or network failures. Moreover , performance takes a major hit as contention on the lock increases. Furthermore, it is neither scalable nor durable.

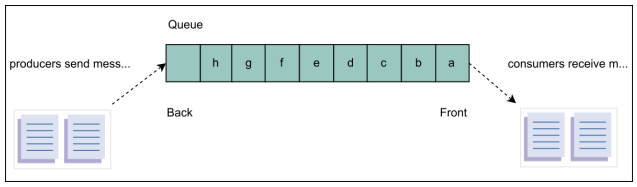


Figure 1.5: Multiple producers and consumers interact via a single messaging queue

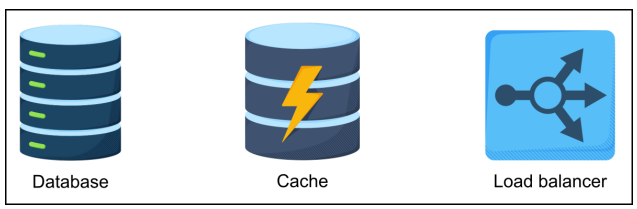
Quiz: Question 1: Can we extend the design of a single-server messaging queue to a distributed messaging queue? **Answer:** A single-server messaging queue has the following drawbacks: 1. High latency: As in the case of a single-server messaging queue, a producer or consumer acquires a lock to access the queue. Therefore, this mechanism becomes a bottleneck when many processes try to access the queue. This increases the latency of the service. 2. Low availability: Due to the lack of replication of the messaging queue,

the producer and consumer process might be unable to access the queue in events of failure. This reduces the system's availability and reliability. 3. Lack of durability: Due to the absence of replication, the data in the queue might be lost in the event of a system failure. 4. Scalability: A single-server messaging queue can handle a limited number of messages, producers, and consumers.

Therefore, it is not scalable. To extend the design of a single-server messaging queue to a distributed messaging queue, we need to make extensive efforts to eliminate the drawbacks outlined above.

2.3 Building blocks we will use

The design of a distributed messaging queue utilizes the following building blocks:



- **Database(s)** will be required to store the metadata of queues and users.
- **Caches** are important to keep frequently accessed data, whether it be data pertaining to users or queue metadata.
- **Load balancers** are used to direct incoming requests to servers where the metadata is stored.

In our discussion on messaging queues, we focused on their functional and non-functional requirements. Before moving on to the process of designing a distributed messaging queue, it's essential for us to discuss some key considerations and challenges that may affect the design.

3 Considerations of a Distributed Messaging Queue's Design

Before embarking on our journey to design a distributed messaging queue, let's discuss some major factors that could significantly affect the design. These include the order of messages, the effect of the ordering on

performance, and the management of concurrent access to the queue. We discuss each of these factors in detail below.

3.1 Ordering of messages

A messaging queue is used to receive messages from producers. These messages are consumed by the consumers at their own pace. Some operations are critical in that they require strict ordering of the execution of the tasks, driven by the messages in the queue. For example, while chatting over a messenger application with a friend, the messages should be delivered in order; otherwise, such communication can be confusing, to say the least. Similarly , emails received by a user from different users may not require strict ordering. Therefore, in some cases, the strict order of incoming messages in the queue is essential, while many use cases can tolerate some reordering.
Let's discuss the following two categories of message ordering in a queue:

- *Best-effort ordering*
- *Strict ordering*

In a queue, the order of messages is implicitly associated with the incoming messages. Once the messages are put in a queue, the same order is followed in the consumption and processing of these messages.
For concurrent producers putting messages in the same queue, the order is not well defined until producers provide order information—for example, timestamps or sequence numbers. Without any ordering information, the queue puts messages in the queue in whatever order they arrive at the service.
For concurrent consumers fetching messages from the same queue, ordering can again become a complicated issue. While the queue can hand over messages one after the other in the same order as they were entered in the queue, two consumers almost concurrently processing two messages might need an application-specific ordering mechanism. The queue might help by tagging a message's ordering information, sequence number or timestamp, while handing out a message from the queue.

Best-effort ordering

With the **best-effort ordering** approach, the system puts the messages in a specified queue in the same order that they're received.
For example, as shown in the following figure, the producer sends four messages, A, B, C, and D, in the same order as illustrated. Due to network congestion or some other issue, message B is received after message D. Hence , the order of messages is A, C, D, and B at the receiving end. Therefore, in this approach, the messages will be put in the queue in the same order they were received instead of the order in which they were produced on the client side.

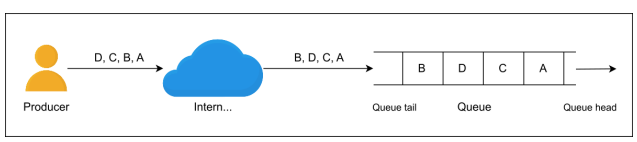


Figure 1.6: Best-effort ordering: Messages are placed in a queue in the same order that they're received and not in the order they were sent

Strict ordering

The strict ordering technique preserves the ordering of messages more rigorously. Through this approach, messages are placed in a queue in the order that they're produced.

Before putting messages in a queue in the correct sequence, it's crucial to have a mechanism to identify the order in which the messages were produced on the client side. Often , a unique identifier or time-stamp is used to mark a message when it's produced.

Quiz: Question 1: Who'll be responsible for providing the sequence numbers? **Answer:** The responsibility for providing sequence numbers lies with the client.

The system facilitates this by offering essential libraries or APIs that the client can integrate into their application. These tools allow the client to assign sequence numbers to messages as they are produced, ensuring each message can be tracked and ordered accurately. This approach helps maintain consistency and enables the system to process and deliver messages in the intended sequence.

One of the following three approaches can be used for ordering incoming messages:

1. **Monotonically increasing numbers:** One way to order incoming messages is to assign monotonically increasing numbers to messages on the server side. When the first message arrives, the system assigns it a number, such as 1. It then assigns the number 2 to the second message, and so on.
   However, there are potential drawbacks to this approach. First , when a burst of requests is received, it acts as a bottleneck that affects the system's performance because the system has to assign an ID in a specified sequence to a message while the other messages wait for their turn.
   Second, it still doesn't tackle the problem that arises when a message is received before the one that's produced earlier at the client side. Because of this, it doesn't guarantee that it will generate the correct order for the messages produced at the client side.

2. **Causality-based sorting at the server side:** Keeping in view the drawbacks of using monotonically increasing numbers, another approach that can be used for time-stamping and ordering of incoming messages is causality-based sorting. In this approach, messages are sorted based on the time stamp that was produced at the client side and are put in a queue accordingly. The major drawback of this approach is that for multiple client sessions, the service can't determine the order in terms of wall-clock time.

3. **Using time stamps based on synchronized clocks:** To tackle the potential issues that arise with both of the approaches described above, we can use another appropriate method to assign time stamps to messages that's based on synchronized clocks. In this approach, the time stamp (ID) provided to each message through a synchronized clock is unique and in the correct sequence of production of messages. We can tag a unique process identifier with the time stamp to make the overall message identifier unique and tackle the situation when two concurrent sessions ask for a time stamp at the exact same time. Moreover, with this approach, the server can easily identify delayed messages based on the time stamp and wait for the delayed messages.
   As we discussed in the section on the sequencer building block, we can get sequence numbers that fulfill double duty as sequence numbers and globally synchronized wall clock time stamps. Using this approach, our service can globally order messages across client sessions as well.

To conclude, the most appropriate mechanism to provide a unique ID or time stamp to incoming messages, from among the three approaches described above, involves the use of synchronized clocks.

Sorting Once messages are received at the server side, we need to sort them based on their time stamps. Therefore, we use an appropriate online sorting(Online sorting means that we need to keep some data sorted as new data keeps coming in.) algorithm for this purpose.

Quiz: Question 1: Suppose that a message sent earlier arrives late due to a network delay.

What would be the proper approach to handle such a situation? **Answer:** The simple solution in such cases is to reorder the queue. Two scenarios can arise from this. First, reordering puts the messages in the correct order. Second, we've already handed out newer messages to the consumers.

If an old message comes when we've already handed out a newer message, we put it in a special queue, and the client handles that situation. The client may later decide whether to consume the message if the message does not affect the intended operation or discard it if it's not needed.

3.2 Effect on performance

Primarily, a queue is designed for first-in, **first-out (FIFO) operations**;. First-in, first-out operations suggest that the first message that enters a queue is always handed out first. However, it isn't easy to maintain this strict order in distributed systems. Since message A was produced before message B, it's still uncertain that message A will be consumed before message B. Using monotonically increasing message identifiers or causality-bearing identifiers provide high throughput while putting messages in a queue. Though the need for the online sorting to provide a strict order takes some time before messages are ready for extraction. To minimize latency caused by the online sorting, we use a **time-window**(We have to sort messages received within a specific time frame and then put them in the relevant queue (the oldest will be the first to be put in the queue).) approach.
Similarly, for strict ordering at the receiving end, we need to serialize all the requests to give out messages one by one. If that's not required, we have better throughput and lower latency at the receiving end.
Due to the reasons mentioned above, many distributed messaging queue solutions either don't guarantee a strict order or have limitations around throughput. As we saw previously, the queues have to perform many additional validations and coordination operations to maintain the order.

Managing concurrency

Concurrent queue access needs proper management. Concurrency can take place at the following stages:

- When multiple messages arrive at the same time.
- When multiple consumers request concurrently for a message.

The first solution is to use the locking mechanism. When a process or thread requests a message, it should acquire a lock for placing or consuming messages from the queue. However, as was discussed earlier, this approach has several drawbacks. It's neither scalable nor performant.
Another solution is to serialize the requests using the system's buffer at both ends of the queue so that the incoming messages are placed in an order, and consumer processes also receive messages in their arrival sequence. By serializing requests, we mean that the requests (either for putting data or extracting data), which come to the server would be queued by the OS, and a single application thread will put them in the queue (we can assume that both kinds of requests, put and extract come to the same port) without any locking. It will be a possible lock-free solution, providing high throughput. This is a more viable solution because it can help us avoid the occurrence of race conditions.
Applications might use multiple queues with dedicated producers and consumers to keep the ordering cost per queue under check, although this comes at the cost of more complicated application logic.

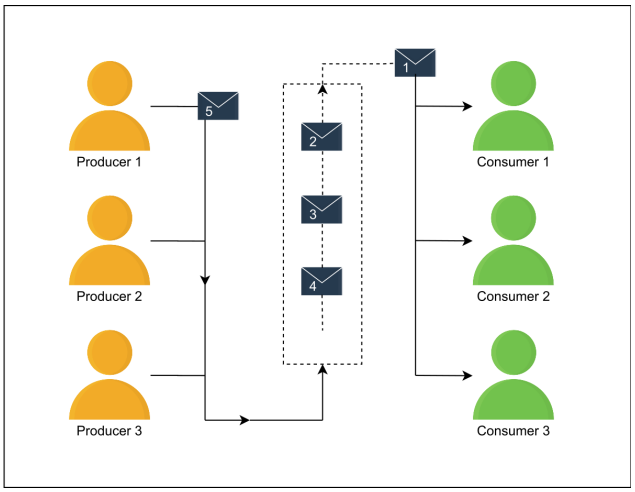


Figure 1.7: Avoiding race conditions: Producers and consumers are serialized at both ends of the queue

In this lesson, we discussed some key considerations and challenges in the design process of a messaging queue and answered the following questions:

- Why is the order of messages important, and how do we enforce that order?
- How does ordering affect performance? How do we handle concurrency while accessing a queue?

Now, we are ready to start designing a distributed messaging queue.

4 Design of a Distributed Messaging Queue: Part 1

So far, we've discussed the requirements and design considerations of a distributed messaging queue. Now , let's begin with learning about the high-level design of a distributed messaging queue.

4.1 Distributed messaging queue

Unlike a single-server messaging queue, a distributed messaging queue resides on multiple servers. A distributed messaging queue has its own challenges. However, it resolves the drawbacks of a single-server messaging queue if designed properly.

The following sections focus on the scalability, availability, and durability issues of designing a distributed messaging queue by introducing us to a more fault-tolerant architecture of a messaging queue.

High-level design

Before diving deep into the design, let's assume the following points to make the discussion more simple and easy to understand. In the upcoming material, we discuss how the following assumptions enable us to eliminate the problems in a single-server solution to the messaging queue.

1. Queue data is replicated using either a primary-secondary or quorum-like system inside a cluster (read through the Data Replication lesson for more details). Our service can use data partitioning if the queue gets too long to fit on a server. We can use a consistent hashing-like scheme for this purpose, or we may use a key-value store where the key might be the sequence numbers of the messages. In that case, each shard is appropriately replicated (refer to the Partition lesson for more details on this).

2. We also assume that our system can auto-expand and auto-shrink the resources as per the need to optimally utilize resources.

The following figure demonstrates a high-level design of a distributed messaging queue that's composed of several components.

The essential components of our design are described in detail below.

Load balancer The load balancer layer receives requests from producers and consumers, which are forwarded to one of the front-end servers. This layer consists of numerous load balancers. Therefore, requests are accepted with minimal latency and offer high availability.

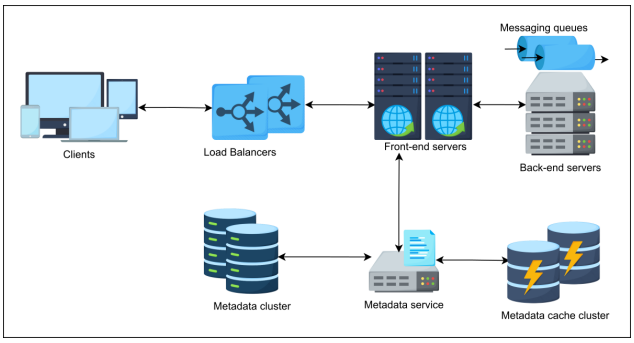


Figure 1.8: High-level architecture of distributed messaging queue

Front-end service The front-end service comprises stateless machines distributed across data centers. The front-end provides the following services:

- **Request validation:** This ensures the validity of a request and checks if it contains all the necessary information.
- **Authentication and authorization:** This service checks if the requester is a valid user and if these services are authorized for use to a requester.
- **Caching:** In the front-end cache, metadata(The metadata of a messaging queue typically includes essential information such as the queue's name, attributes, configuration settings, and may also encompass details like message retention policies, access permissions, and queue-specific metrics. It serves as a crucial reference for managing the behavior and characteristics of the messaging queue within a distributed system.) information is stored related to the frequently used queues. Along with this, user-related data is also cached here to reduce request time to authentication and authorization services.
- **Request dispatching:** The front-end is also responsible for calling two other services, the back-end and the metadata store. Differentiating calls to both of these services is one of the responsibilities of the front-end.
- **Request deduplication:** The front-end also tracks information related to all the requests, therefore, it also prevents identical requestsDeduplication from being put in a queue. Deciding what to do about duplicates might be as easy as searching a hash key in a store. If something is found in the store, this implies a duplicate and the message can be rejected.
- **Usage data collection:** This refers to the collection of real-time data that can be used for audit purposes.

Metadata service This component is responsible for storing, retrieving, and updating the metadata of queues in the metadata store(A cluster of persistent data storage responsible for storing metadata related to all the queues available in the system.) and cache(A cluster of metadata, frequently accessed whenever clients' requests arrive.). Whenever a queue is created or deleted, the metadata store and cache are updated accordingly. The metadata service acts as a middleware between the front-end servers and the data layer. Since the metadata of the queues is kept in the cache, the cache is checked first by the front-end servers for any relevant information related to the receipt of the request. If a cache miss occurs, the information is retrieved from the metadata store and the cache is updated accordingly.
There are two different approaches to organizing the metadata cache clusters:

1. If the metadata that needs to be stored is small and can reside on a single machine, then it's replicated on each cluster server. Subsequently , the request can be served from any random server. In this approach, a load balancer can also be introduced between the front-end servers and metadata services.

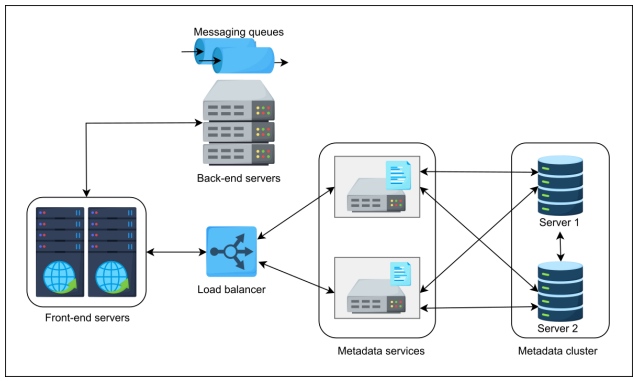


Figure 1.9: A load balancer is introduced between the front-end servers and metadata services

2. If the metadata that needs to be stored is too large, then one of the following modes can be followed:

- The first strategy is to use the sharding approach to divide data into different shards. **Sharding** can be performed based on some partition key or hashing techniques, as was discussed in the lesson on database partitioning. Each shard is stored on a different host in the cluster. Moreover, each shard is also replicated on different hosts to enhance availability. In this cluster-organization approach, the front-end server has a mapping table between shards and the hosts. Therefore, the front-end server is responsible for redirecting requests to the host where the data is stored.

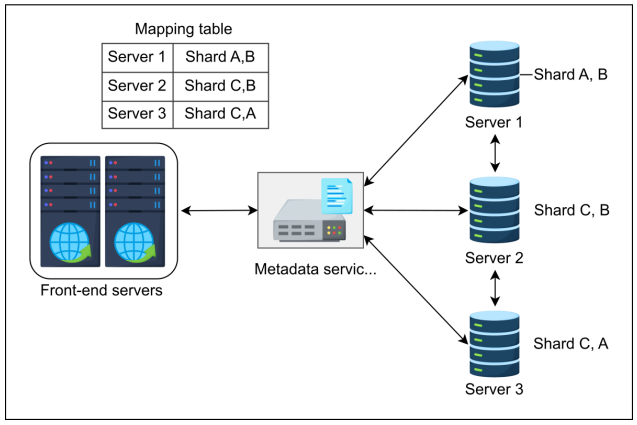


Figure 1.10: Mapping table resides on the front-end servers

- The second approach is similar to the first one. However, the mapping table in this approach is stored on each host instead of just on the front-end servers. Because of this, any random host can receive a request and forward it to the host where the data resides. This technique is suitable for read-intensive applications.

In our discussion on distributed messaging queues, we focused on the high-level design of this type of queue. Furthermore , we explored each component in the high-level design, including the following:

- Front-end servers and the services required of them for smooth operations.
- Load balancers.
- Metadata services.
- Metadata clusters and their organization.

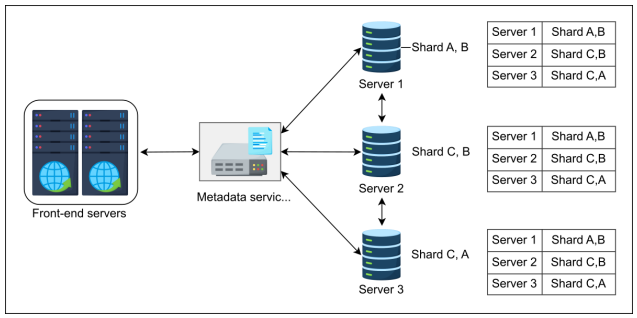


Figure 1.11: Mapping table resides on each metadata server

A vital part of the design is the organization of servers at the back-end for queue creation, deletion, and other such operations. The next lesson focuses on the organization of back-end servers and the management of queues, along with other important operations related to message delivery and retrieval.

5 Design of a Distributed Messaging Queue: Part 2

In the previous lesson, we discussed the responsibilities of front-end servers and metadata services. In this lesson, we'll focus on the main part of the design where the queues and messages are stored: the back-end service.

5.1 Back-end service

This is the core part of the architecture where major activities take place. When the front-end receives a message, it refers to the metadata service to determine the host where the message needs to be sent. The message is then forwarded to the host and is replicated on the relevant hosts to overcome a possible availability issue. The message replication in a cluster on different hosts can be performed using one of the following two models:

1. *Primary-secondary model*
2. *A cluster of independent hosts*

Before delving into the details of these models, let's discuss the two types of cluster managers responsible for queue management: *internal* and *external* cluster managers. The differences between these two cluster managers are shown in the following table.

Internal versus External Cluster Managers

Internal Cluster Manager	**External Cluster Manager**
It manages the assignment of queues within a cluster.	It manages the assignment of queues across clusters.
It knows about each and every node within a cluster.	It knows about each cluster. However, it doesn't have information on every host that's present inside a cluster.
It listens to the heartbeat from each node.	It monitors the health of each independent cluster.
It manages host failure, instance addition, and removals from the cluster.	It manages and utilizes clusters.
It partitions a queue into several parts and each part gets a primary server.	It may split a queue across several clusters, so that messages for the same queue are equally distributed between several clusters.

Table 1.1: Internal versus External Cluster Managers

Primary-secondary model

In the **primary-secondary model**, each node is considered a **primary host** for a collection of queues. The responsibility of a primary host is to receive requests for a particular queue and be fully responsible for data replication. The request is received by the front-end, which in turn communicates with the metadata service to determine the primary host for the request.
For example, suppose we have two queues with the identities 101 and 102 residing on four different hosts A, B, C, and D. In this example, instance B is the primary host of queue 101 and the secondary hosts where the queue 101 is replicated are A and C. As the front-end receives message requests, it identifies the primary server from the internal cluster manager through the metadata service. The message is retrieved from the primary instance, which is also responsible for deleting the original message upon usage and all of its replicas.
As shown in the following illustration, the internal cluster manager is a component that's responsible for mapping between the primary host, secondary hosts, and queues. Moreover , it also helps in the primary host selection. Therefore, it needs to be reliable, scalable, and performant.

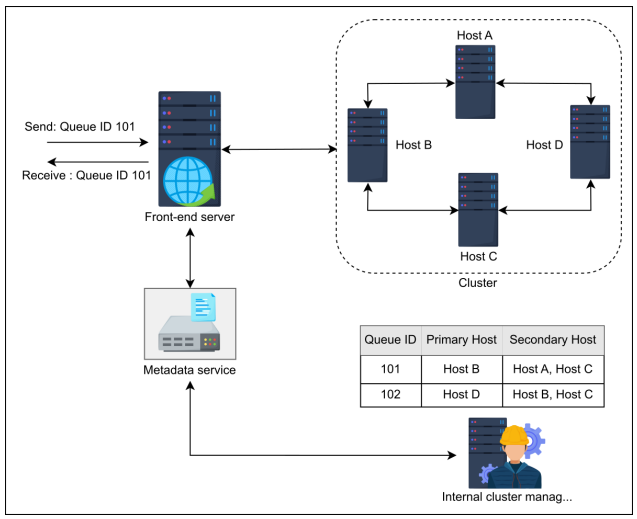


Figure 1.12: Primary-secondary model of distributed queue: A request is received for a queue with ID 101, which is served accordingly

A cluster of independent hosts

In the approach involving a **cluster of independent hosts**, we have several clusters of multiple independent hosts that are distributed across data centers. As the front-end receives a message, it determines the corresponding cluster via the metadata service from the external cluster manager. The message is then forwarded to a random host in the cluster, which replicates the message in other hosts where the queue is stored.

> **Quiz: Question 1:** How does a random host within a cluster replicate data—that is,
>
> messages—in the queues on other hosts within the same cluster? **Answer:** Each host consists of mapping between the queues and the hosts within a cluster, making the replication easier. Assume that we have a cluster,
>
> say Y, having hosts A, B, and C. This cluster has two queues with IDs 101 and 103 stored on different hosts, as shown in the following table. This table is stored on each host within the cluster Y. When a random host receives a message, say host C, for a queue having ID 103, host C replicates this message on the other hosts where the queue 103 is stored, i.e., Node A and Node B.

The same process is applied to receive message requests from the consumer. Similar to the first approach, the randomly selected host is responsible for message delivery and cleanup upon a successful processing of

the message.
Furthermore, another component called an **external cluster manager** is introduced, which is accountable for maintaining the mapping between queues and clusters, as shown in the following figure. The external cluster manager is also responsible for queue management and cluster assignment to a particular queue.
The following figure illustrates the cluster of independent hosts. There are two clusters, A and B, which consist of several nodes. The external cluster manager has the mapping table between queues and their corresponding cluster. Whenever a front-end receives a request for a queue, it determines the corresponding cluster for the queue and forwards the request to the cluster where the queue resides. The nodes within that cluster are responsible for storing and sending messages accordingly.

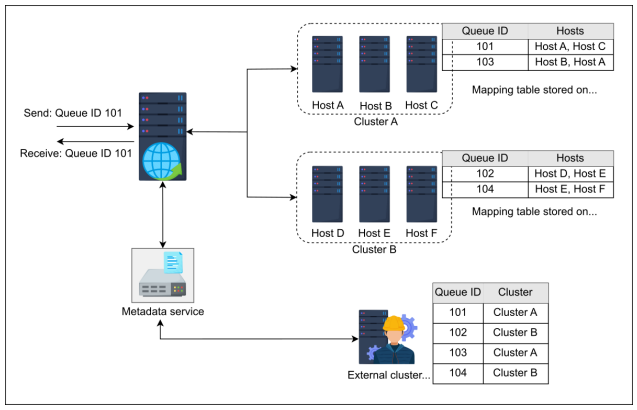


Figure 1.13: A cluster of independent hosts that consist of distributed queues

Quiz: Question 1: How is message replication handled in distributed queue systems? **Answer:** There are two ways to replicate messages in a queue residing on multiple hosts. 1. Synchronous replication 2. Asynchronous replication In synchronous replication, the primary host is responsible for replicating the message in all the relevant queues on other hosts. After acknowledgment from secondary hosts, the primary host then notifies the client regarding the reception of messages. In this approach, messages remain consistent in all queue replicas; however, it costs extra delay in communication and causes partial to no availability while an election is in progress to promote a secondary as a primary. In asynchronous replication, once the primary host receives the messages, it acknowledges the client and, in the next step, starts replicating the message to other hosts. This approach has other problems, such as replication lag and consistency issues. Based on the needs of an application, we can pick one or the other.

We have completed the design of a distributed messaging queue and discussed the two models for organizing the back-end server. We also described the management process of queues and how messages are processed at the back-end. Furthermore , we discussed how back-end servers are managed through different cluster managers.
In the next lesson, we'll discuss how our system fulfills the functional and non-functional requirements that were described earlier in the chapter.

6 Evaluation of a Distributed Messaging Queue's Design

We completed the process of designing a distributed messaging queue. Now , let's analyze whether the design met the functional and non-functional requirements of a distributed messaging queue.

6.1 Functional requirements compliance

- **Queue creation and deletion:** When a request for a queue is received at the front-end, the queue is created with all the necessary details provided by the client after undergoing some essential checks. The corresponding cluster manager assigns servers to the newly created queue and updates the information in the metadata stores and caches through a metadata service.
  Similarly, the queue is deleted when the client doesn't need it anymore. The responsible cluster manager deallocates the space occupied by the queue and, consequently, deletes the data from all the metadata stores and caches.

  Quiz: Question 1: How do we handle messages that can't be processed—here meaning consumed—after maximum processing attempts by the consumer? **Answer:** A special type of queue, called a dead-letter queue, can be provided to handle messages that aren't consumed after the maximum number of processing attempts have been made by the consumer. This type of queue is also used for keeping messages that can't be processed successfully due to the following factors: The messages intended for a queue that doesn't exist anymore. The queue length limit is exceeded, although this would rarely occur with our current design. The message expires due to per-message time to live (TTL). A dead-letter queue is also important for determining the cause of failure and for identifying faults in the system.

- **Send and receive messages:** Producers can deliver messages to specific queues once they are created. At the back-end, receiving messages are sorted based on time stamps to preserve their order and are placed in the queue. Similarly, a consumer can retrieve messages from a specified queue.

When a message is received from a producer for a specific queue, the front-end identifies the primary host or cluster, depending on the replication model, where the queue resides. The request is then forwarded to the corresponding entity(To a primary node in the primary-secondary model, or to a host in a cluster of independent nodes model.) and put in the queue.

- **Message deletion:** Primarily, two options are used to delete a message from a queue.

1. The first option is not to delete a message after it's consumed. However, in this case, the consumer is responsible for keeping track of what's consumed. For this, we need to maintain the order of messages in the queue and keep track of a message within a queue. A job can then delete the message when the expiration conditions are met. Apache Kafka mostly uses this idea where multiple processes can consume a message.

2. The second approach also doesn't delete a message after it's consumed. However, it's made invisible for some time via an attribute—for example, visibility_timeout. This way, the other consumers are unable to get messages that have already been consumed. The message is then deleted by the consumer via an API call.

In both cases, the message being retrieved by the consumer is only deleted by the consumer. The reason behind this is to provide high durability if a consumer can't process a message due to some failure. In such a case, in the absence of a delete call, the consumer can retrieve the message again when it comes back.
Moreover, this approach also provides at-least-once delivery semantic(In this type of delivery semantic, a message can be delivered more than once. Most distributed systems implement this type of delivery semantic because it is easier to implement. An underlying assumption is that the application will take care of possible redundant deliveries and processing (usually, duplicates do not hurt an application because processing is idempotent).). For example, when a worker fails to process the message, another worker can retrieve the message after it becomes visible in the queue.

Quiz: Question 1: What happens when the visibility timeout of a specific message expires and the consumer is still busy processing the message? **Answer:** The message becomes visible, and another worker can receive the message,

thereby duplicating the processing. To avoid such a situation, we ensure that the application sets a safe threshold for visibility timeout.

6.2 Non-functional compliance

- **Durability:** To achieve durability, the queues' metadata is replicated on different nodes. Similarly, when a message is received, it's replicated in the queues that reside on different nodes. Therefore, if a node fails, other nodes can be used to deliver or retrieve messages.

- **Scalability:** Our design components, such as front-end servers, metadata servers, caches, back-end clusters, and more, are horizontally scalable. We can add to or remove their capacity to match our needs. The scalability can be divided into two dimensions:

  - **Increase in the number of messages:** When the number of messages touches a specific limit—say, 80%—the specified queue is expanded. Similarly, the queue is shrunk when the number of messages drops below a certain threshold.
  - **Increase in the number of queues:** With an increasing number of queues, the demand for more servers also increases, in which case the cluster manager is responsible for adding extra servers. We commission nodes so that there is performance isolation between different queues. An increased load on one queue shouldn't impact other queues.

- **Availability:** Our data components, metadata and actual messages, are properly replicated inside or outside the data center, and the load balancer routes traffic around failed nodes. Together, these mechanisms make sure that our system remains available for service under faults.

- **Performance:** For better performance we use caches, data replication, and partitioning, which reduces the data reads and writes time. Moreover, the best effort ordering strategy for ordering messages is there to use to increase the throughput and lower the latency when it's necessary. In the case of strict ordering, we also suggest time-window based sorting to potentially reduce the latency.

Describe how a distributed messaging queue would be used in an online multiplayer game to manage real-time player actions (movement, attacks, etc.). What are the crucial requirements for this scenario?

Question:
How can you use a distributed messaging queue in online games?
Reference Answer:
Distributed messaging queues enable real-time gameplay by:

- Collecting player actions: Movement, attacks, etc., are captured and turned into messages.

- Enqueuing and distributing messages: Messages are buffered and sent to relevant game servers/clients.

- Updating game state: Servers/clients process messages in FIFO and update the game world accordingly.

Requirements: Low latency, potentially relaxed ordering, high availability, and ability to handle bursts of messages during peak usage, reliability.
Additional Info: " Messaging queues help in online multiplayer gaming systems by: - Decoupling components: Systems work independently for improved resilience. - Synchronizing data: Ensures consistency across gaming servers. - Maintaining order: Guarantees actions occur in the correct sequence. "

6.3 Conclusion

We discussed many subtleties in designing a FIFO queue in a distributed setting. We saw that there is a trade-off between strict message production, message extraction orders, and achievable throughput and latency. Relaxed ordering gives us a higher throughput and lower latency. Asking for strict ordering forces the system to do extra work to enforce wall-clock or causality-based ordering. We use different data stores with appropriate replication and partitioning to scale with data. This design exercise highlights that a construct, a producer-consumer queue, that's simple to realize in a single-OS based system becomes much more difficult in a distributed setting.

7 Quiz on the Distributed Messaging Queue's Design

Quiz

Question 1: What kind of messages do dead-letter queues contain?

- The messages that have been consumed successfully.
- **[CORRECT]** The messages that failed to be consumed and have reached the maximum attempts limit.

  Explanation: A dead letter cannot be delivered or returned because it lacks appropriate directions. Similarly , a queue named after the `dead-letter` contains those messages that have failed to be consumed after the maximum attempts limit.
- The messages that have been produced by a process that's dead now.
- None of the above

Question 2: The purpose of replicating queues on multiple servers is to enhance the _____ of the system.

- security
- consistency
- **[CORRECT]** availability

  Explanation: The purpose of replicating the queues and corresponding messages on multiple nodes is to increase the system's availability. When one node is down, another stand-by server can serve the request.
- None of the above

Question 3: In the **cluster of independent hosts** model, which component is responsible for replicating messages in the other nodes?

- The front-end server
- The cluster manager
- **[CORRECT]** Any random host within the cluster

  Explanation: In the **cluster of independent hosts** model, a random host can receive the message responsible for replicating messages in other hosts in the respective queues.
- None of the above

Question 4: The purpose of a metadata service is _____.

- to store messages in the queues along with their metadata
- **[CORRECT]** to store, retrieve, and update the metadata of queues in the metadata store and cache
  Explanation: Metadata service acts as a middleware between the front-end and metadata storages (and caches). It is responsible for storing, retrieving, and updating the metadata of queues in the metadata stores and caches.
- to provide security features in accessing the queues
- None of the above

Question 5: Which of the following is responsible for partitioning the queue and assigning a primary node to each partition?

- **[CORRECT]** The internal cluster manager
  Explanation: The internal cluster manager partitions a queue into multiple parts and assigns a primary node to each partition. In contrast, the external cluster manager assigns clusters to each queue partition.
- The external cluster manager
- The primary node
- One of the secondary nodes